The Islamia University of Bahawalpur

Department of Information Technology

SOFTWARE REQUIREMENTS SPECIFICATION

(SRS DOCUMENT)

for

AI Expression Detection (Website)

Version 1.0

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ADP INFORMATION TECHNOLOGY

Table of Contents

Version 1.0 1
Application Evaluation History 4
Introduction 3
Purpose 3
Scope 3
Overall description 4
Product perspective 4
Design and implementation constraints 4
Requirement identifying technique 4
Use case diagram 5
Use case description 6
Functional Requirement 12
Non-Functional Requirements 14
Usability: 14
Performance: 14
Reliability: 14
Security: 14
Compatibility: 14
References: 15

Revision History

Name	Date	Reason for changes	Version

Application Evaluation History

Comments (by committee) ***include the ones given at scope time both in doc and presentation**	**Action Taken**

Supervised by:

MAM KAINAT

Signature_____

Introduction

The **Introduction** section provides a comprehensive overview to help stakeholders understand the structure, purpose, and importance of this Software Requirements Specification (SRS) document. It serves as a foundational guide for all stakeholders, including developers, testers, designers, project managers, and clients, offering clarity on the project's objectives, scope, and intended functionalities.

This project introduces a web-based **AI facial expression** and demographic detection system, utilizing the Deep Face library to accurately predict **emotions**, **age**, **gender**, and race from uploaded images. Built with a Django backend and a user-friendly interface, it addresses limitations of existing solutions, offering high accuracy and scalability. Leveraging advanced models like Arc Face and Retina Face, the system is designed for applications in education, healthcare, security, and marketing, with potential for future enhancements like real-time analysis and API integration.
This SRS document is organized to facilitate a clear understanding of the project requirements and deliverables. It defines the functional and non-functional requirements, performance standards, and design constraints that the system must adhere to. Additionally, it includes use case descriptions, diagrams, and specific scenarios to provide a holistic view of the system's behavior.

Purpose

The AI Expression Detection Website is a web-based application designed to analyze facial expressions and demographic attributes (emotion, age, gender, race) from uploaded images using the Deep Face library. The system aims to provide accurate, user-friendly, and scalable facial analysis for applications in education, healthcare, security, and marketing.
It is intended to serve both technical and non-technical users by offering a clean interface and automated backend processing. The platform emphasizes privacy, ensuring images are processed securely and not stored unnecessarily. Additionally, its modular framework allows seamless integration with future technologies and expanding datasets.

Scope

The AI Expression Detection Website enables users to upload images and receive detailed analysis of facial expressions and demographic attributes. Built with Django for the backend and a responsive frontend, the system leverages advanced machine learning models (Deep Face, Arc Face, Retina Face) for robust predictions. It includes a modular design for scalability and future enhancements like real-time analysis and API integration.

Major Features:

- **Image Upload and Analysis:** Users can upload images for facial attribute detection.
- **User-Friendly Interface:** A responsive web interface for easy interaction.
- **Backend Processing**: Django-based backend for efficient image processing and prediction.
- **Database Storage:** Stores user data and analysis history.

- **Scalability:** Modular design for future enhancements (e.g., real-time analysis, API integration).

The system aims to address challenges in existing facial analysis solutions by improving accuracy, usability, and accessibility.

Overall description

Product perspective

The AI Expression Detection Website is a new product designed to provide a robust and accessible platform for facial expression and demographic analysis. Unlike existing systems like VGG-Face or FaceNet, which focus on specific features or lack user-friendliness, this system integrates multiple attribute detection (emotion, age, gender, race) with a scalable, web-based architecture. It is a standalone application with potential for future integration with APIs or real-time video analysis.

Design and implementation constraints

Design Constraints

- **Framework:** Must use Django for backend development.
- **UI Design:** Interface must be simple and responsive.
- **Architecture:** Must follow the MVC (Model-View-Controller) pattern (Django uses MTV, which aligns with MVC principles).
- **Facial Analysis:** Must use the Deep Face library.
- **Database:** Must use PostgreSQL to store user data securely.
- Must support both desktop and mobile devices.

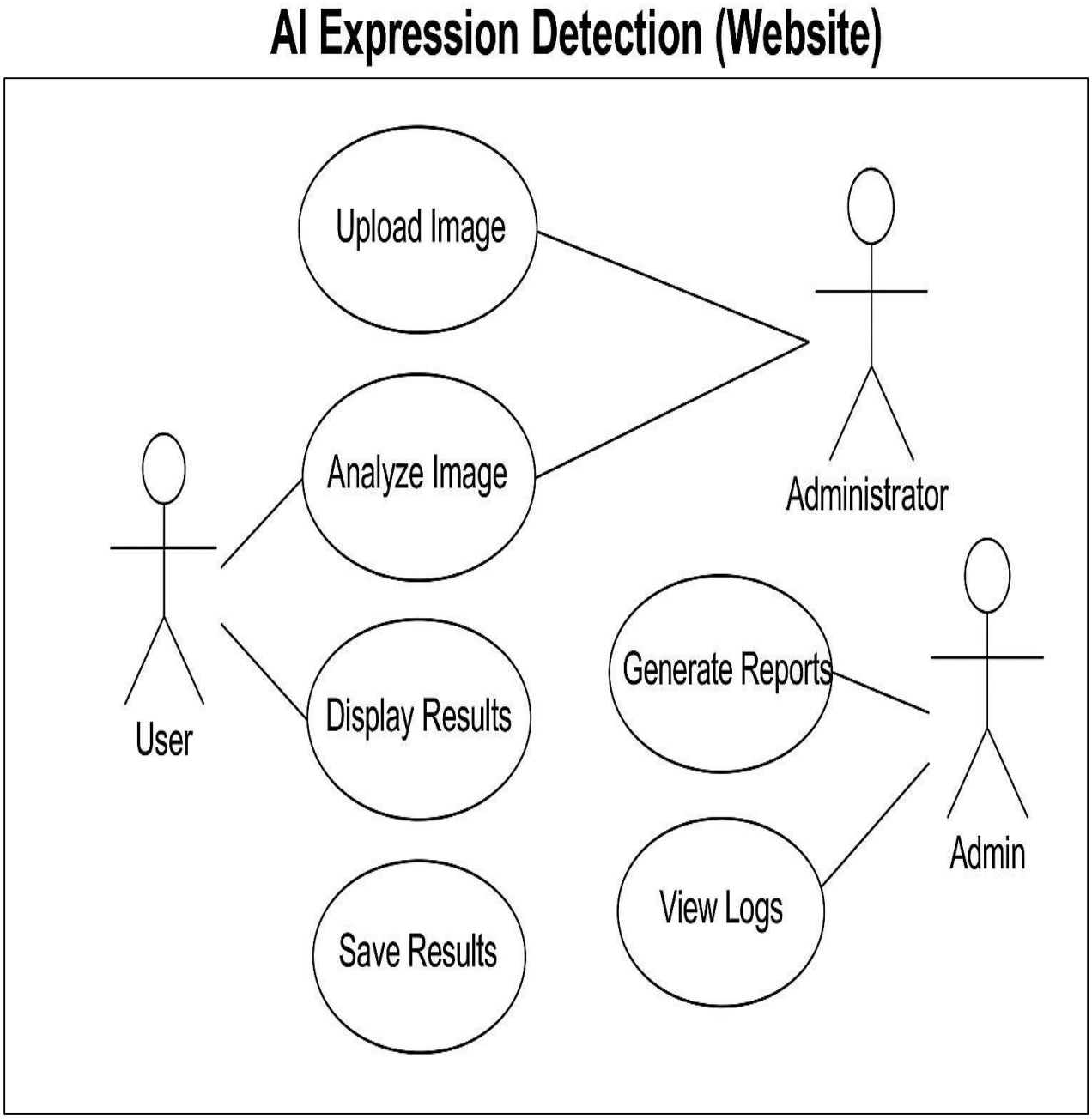
Implementation Constraints

- **Programming Language:** Must use Python 3.8.
- **Development Environment**: Must be hosted on Windows 11.
- **Libraries:** Only open-source Python libraries allowed for facial recognition and analysis.
- **Frontend:** Must be built using HTML5, CSS3, and Bootstrap for styling.
- Internet access is required initially to download Deep Face models.

Requirement identifying technique

For the AI Expression Detection Website, the technique chosen to identify requirements is the **Use Case technique**. This approach is particularly effective for interactive admin role to interact with the system to perform specific tasks.

Use case diagram



Use case description

Upload Image

Use Case ID:	UC-1
Use Case Name:	Upload Image
Actors:	Primary Actor: User
Description:	A User uploads an image to the AI Expression Detection Website for facial attribute analysis (emotions, age, gender, race).
Trigger:	The User clicks the "Upload" button after selecting an image file.
Preconditions:	PRE-1: User is logged into the system. PRE-2: The image is in a supported format (JPEG or PNG)
Postconditions:	POST-1: The image is successfully uploaded and stored temporarily for analysis. POST-2: The system proceeds to analyze the image.
Normal Flow:	1. User navigates to the image upload page. 2. User clicks the "Choose File" button. 3. User selects an image from their device. 4. User clicks the "Upload" button. 5. System validates the image format (see E1). 6. System confirms successful upload and triggers the analysis process.
Alternative Flows: [Alternative Flow 1 – Not in Network]	Alternative Flow 1 – Cancel Upload At step 4 of the normal flow, if the User decides not to proceed: 1. User clicks the "Cancel" button instead of "Upload". 2. System clears the selected image. 3. System returns to the upload page (return to step 1).
Exceptions:	E1: Invalid image format 1. System displays error: "Please upload a JPEG or PNG image." 2a. User selects another image (return to step 3). 2b. Else, User cancels the upload process, and the system terminates the use case.
Business Rules:	BR-1: Only JPEG and PNG images are supported. BR-2: Maximum file size is 5MB.
Assumptions:	Assume 80% of users will upload JPEG images.

Analyze Image

Use Case ID	UC-2
Use Case Name	Analyze Image
Actors	Primary Actor: User
Description	The system analyzes the uploaded image using the Deep Face library to detect facial attributes (emotions, age, gender, race).
Trigger	The system automatically initiates analysis after a successful image upload (triggered by UC-1).
Preconditions	PRE-1: An image has been successfully uploaded (from UC-1). PRE-2: Deep Face library is operational.
Postconditions	POST-1: Analysis results (emotions, age, gender, race) are generated. POST-2: Results are ready to be displayed.
Normal Flow	1. System retrieves the uploaded image. 2. System processes the image using Deep Face library. 3. System applies Arc Face and Retina Face models for accurate detection. 4. System generates analysis results within 5 seconds (see E1). 5. System stores results temporarily for display.
Alternative Flows	**Alternative Flow 1 – Multiple Faces Detected** At step 3 of the normal flow, if multiple faces are detected: 1. System analyzes all detected faces. 2. System generates separate results for each face (e.g., "Face 1: Emotion: Happy, Age: 30", "Face 2: Emotion: Sad, Age: 25"). 3. Return to step 4 of the normal flow.
Exceptions	E1: Analysis takes longer than 5 seconds 1. System displays a timeout error: "Analysis took too long, please try again." 2a. User retries by uploading another image (return to UC-1). 2b. Else, User cancels the process, and the system terminates the use case. E2: No face detected in the image 1. System displays error: "No face detected in the image." 2a. User uploads a different image (return to UC-1). 2b. Else, User cancels the process, and the system terminates the use case.
Business Rules	BR-3: Analysis must complete within 5 seconds. BR-4: At least one face must be detected for analysis to proceed.
Assumptions	Assume 90% of images will contain at least one detectable face.

Display Results

Use Case ID	UC-3
Use Case Name	Display Results
Actors	Primary Actor: User
Description	The system displays the analysis results (emotions, age, gender, race) to the User after processing the image.
Trigger	The system completes image analysis (triggered by UC-2).
Preconditions	PRE-1: Analysis results are generated (from UC-2). PRE-2: User is still on the results page.
Postconditions	POST-1: Results are displayed to the User. POST-2: User can view and interact with the results.
Normal Flow	1. System retrieves the analysis results. 2. System formats results (e.g., "Emotion: Happy, Age: 30, Gender: Female, Race: Asian"). 3. System displays results on the webpage. 4. User views the results.
Alternative Flows	**Alternative Flow 1 – Request Detailed View** At step 4 of the normal flow, if the User wants more details: 1. User clicks a "Show Details" button. 2. System displays additional analysis data (e.g., confidence scores for each attribute, such as "Emotion: Happy (Confidence: 92%)"). 3. User views the detailed results.
Exceptions	E1: User navigates away before results are displayed 1. System terminates the display process. 2. Results are still saved (proceeds to UC-4).
Business Rules	BR-5: Results must be displayed in a clear, readable format. BR-6: Results page must load within 3 seconds.
Assumptions	Assume 95% of users will wait to view the results after analysis.

Save Results

Use Case ID	UC-4
Use Case Name	Save Results
Actors	Primary Actor: User
Description	The system saves the analysis results and associated user data in the database for future reference.
Trigger	The system completes displaying the results (triggered by UC-3).
Preconditions	PRE-1: Analysis results are generated and displayed (from UC-3). PRE-2: Database is accessible.
Postconditions	POST-1: Results are stored in the database. POST-2: User can access past results in their history.
Normal Flow	1. System retrieves the analysis results. 2. System associates results with the User's account. 3. System saves the results in the database (SQLite/PostgreSQL). 4. System logs the transaction for future retrieval.
Alternative Flows	**Alternative Flow 1 – User opts Out of Saving** At step 2 of the normal flow, if the User has disabled saving in their settings: 1. System skips saving the results to the database. 2. System logs the transaction as "Not Saved" for auditing purposes. 3. Use case terminates.
Exceptions	E1: Database connection failure 1. System displays error: "Failed to save results, please try again later." 2a. System retries saving (return to step 3). 2b. Else, results are discarded, and the use case terminates.
Business Rules	BR-7: Results must be saved within 2 seconds of display. BR-8: Each result entry must include a timestamp and User ID.
Assumptions	Assume 99% of save operations will succeed under normal server conditions.

Generate Reports

Use Case ID	UC-5
Use Case Name	Generate Reports
Actors	Primary Actor: Administrator
Description	The Administrator generates reports summarizing analysis results and system usage statistics.
Trigger	The Administrator clicks the "Generate Report" button on the admin dashboard.
Preconditions	PRE-1: Administrator is logged into the system with appropriate permissions. PRE-2: Analysis results are available in the database.
Postconditions	POST-1: A report is generated and displayed to the Administrator. POST-2: The report can be downloaded or viewed on the dashboard.
Normal Flow	Administrator navigates to the reporting section of the admin dashboard. Administrator selects the report type (e.g., usage statistics, analysis summary). Administrator clicks the "Generate Report" button. System retrieves relevant data from the database. System generates the report within 10 seconds. System displays the report to the Administrator.
Alternative Flows	Alternative Flow 1 – Export Report At step 6 of the normal flow, if the Administrator wants to export the report: Administrator clicks the "Export" button. System exports the report as a PDF or CSV file. Administrator downloads the file.
Exceptions	E1: Report generation takes longer than 10 seconds System displays a timeout error: "Report generation took too long, please try again." 2a. Administrator retries generating the report (return to step 3) . 2b. Else, Administrator cancels the process, and the system terminates the use case.
Business Rules	BR-9: Reports must be generated within 10 seconds.
Assumptions	Assume 80% of Administrators will generate usage statistics reports.

View Logs

Use Case ID	UC-6
Use Case Name	View Logs
Actors	Primary Actor: Admin
Description	The Admin views system logs to monitor user activities and system performance.
Trigger	The Admin clicks the "View Logs" button on the admin dashboard.
Preconditions	PRE-1: Admin is logged into the system with appropriate permissions. PRE-2: Logs are available in the system.
Postconditions	POST-1: Logs are displayed to the Admin. POST-2: Admin can filter or search through the logs.
Normal Flow	Admin navigates to the logs section of the admin dashboard. Admin clicks the "View Logs" button. System retrieves logs from the database. System displays logs within 5 seconds. Admin views the logs.
Alternative Flows	Alternative Flow 1 – Filter Logs At step 5 of the normal flow, if the Admin wants to filter logs: Admin selects a filter (e.g., by date, user, or activity type). System updates the displayed logs based on the filter. Admin views the filtered logs.
Exceptions	E1: Log retrieval takes longer than 5 seconds 1. System displays a timeout error: "Log retrieval took too long, please try again." 2a. Admin retries viewing the logs (return to step 2). 2b. Else, Admin cancels the process, and the system terminates the use case.
Business Rules	BR-10: Logs must be accessible within 5 seconds.
Assumptions	Assume 90% of Admins will view logs to monitor user activities.

Functional Requirement

Identifier	FR-1.1
Title	Image Upload
Requirement	The User shall be able to upload an image (JPEG or PNG) to the system for facial attribute analysis.
Source	Frontend Module
Rationale	Enables the core functionality of providing images for analysis, which is the primary purpose of the system.
Business Rule	BR-1: Only JPEG and PNG images are supported. BR-2: Maximum file size is 5MB.
Dependencies	None
Priority	High

Identifier	FR-2.1
Title	Image Analysis
Requirement	The system shall analyze the uploaded image using the Deep Face library to detect facial attributes (emotions, age, gender, race) within 5 seconds.
Source	Backend Module
Rationale	Provides the primary functionality of detecting facial attributes, which is the main purpose of the system.
Business Rule	BR-3: Analysis must complete within 5 seconds. BR-4: At least one face must be detected for analysis to proceed.
Dependencies	FR-1 (Image must be uploaded before analysis can occur).
Priority	High

Identifier	FR-3.1
Title	Display Analysis Results
Requirement	The system shall display the analysis results (emotions, age, gender, race) to the User in a clear and readable format within 3 seconds of analysis completion.
Source	Frontend Module
Rationale	Ensures that users can view the results of the analysis in a timely and understandable manner.
Business Rule	BR-5: Results must be displayed in a clear, readable format. BR-6: Results page must load within 3 seconds.
Dependencies	FR-2 (Analysis results must be generated before they can be displayed).
Priority	High

Identifier	FR-4.1
Title	Store Analysis Results
Requirement	The system shall save the analysis results (emotions, age, gender, race) with the User ID and timestamp in the database after display.
Source	Backend Module
Rationale	Enables tracking of analysis history for users and system performance monitoring.
Business Rule	BR-7: Results must be saved within 2 seconds of display. BR-8: Each result entry must include a timestamp and User ID.
Dependencies	FR-3.1
Priority	Medium

Identifier	FR-5.1
Title	Generate Reports
Requirement	The Administrator shall be able to generate reports summarizing analysis results and system usage statistics.
Source	Backend Module
Rationale	Allows Administrators to monitor system performance and user activity for decisionmaking.
Business Rule	Reports must be generated within 10 seconds.
Dependencies	FR-4.1
Priority	Medium

Identifier	FR-6.1
Title	View Logs
Requirement	The Admin shall be able to view system logs to monitor user activities and system performance.
Source	Backend Module
Rationale	Provides Admins with insights into system operations and user interactions for troubleshooting and auditing.
Business Rule	Logs must be accessible within 5 seconds.
Dependencies	None
Priority	Medium

Non-Functional Requirements

Usability:

1. The system shall allow users to upload an image and view results in no more than three clicks, taking under 6 second's total.
2. The interface shall adapt to screen sizes from 320px to 1920px with no layout errors, ensuring responsive design.
3. The system shall provide error messages (e.g., for invalid image formats) in clear language, with retry options visible.

Performance:

1. The system shall analyze an image using the Deep Face library within 5 seconds on a server with 8GB RAM and 2.4GHz CPU.
2. The system shall support 500 concurrent users with page load times under 3 seconds over a 10 Mbps connection.
3. The database shall handle 5,000 analysis records with query response times under 1 second.

Reliability:

1. The system shall achieve 99.9% uptime over 30 days, excluding scheduled maintenance windows.
2. The system shall successfully process 90% of images despite challenging conditions (e.g., low lighting, occlusions).
3. The system shall recover from analysis failures within 30 seconds, prompting users to retry with a new image.

Security:

1. The system shall encrypt image uploads and results in transit using HTTPS with TLS 1.3.
2. User data (e.g., analysis results, User ID) shall be stored in the database using AES-256 encryption.
3. The system shall require user authentication for image uploads, using session tokens valid for 24 hours.

Compatibility:

1. The system shall support the latest versions of Google Chrome, Firefox (100+), Microsoft Edge, and Safari (15.0+).
2. The system shall function on devices with iOS 15+ and Android 12+, ensuring mobile responsiveness.

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